

Grid Interactive Solar PV Based Water Pumping Using BLDC Motor Drive

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Abstract—This paper proposes a bidirectional power flow control of a grid interactive solar photovoltaic (PV) fed water pumping system. A brushless DC (BLDC) motor-drive without phase current sensors, is used to run a water pump. This system enables a consumer to operate the water pump at its full capacity for 24-hours regardless of the climatic condition and to feed a single phase utility grid when the water pumping is not required. The full utilization of a PV array and motor-pump is made possible in addition to an enhanced reliability of the pumping system. A single phase voltage source converter (VSC) with a unit vector template (UVT) generation technique accomplishes a bidirectional power flow control between the grid and the DC bus of voltage source inverter (VSI), which feeds a BLDC motor. The VSI is operated at fundamental frequency, which minimizes the switching loss. The maximum power point (MPP) operation of a PV array, and power quality improvements such as power factor correction and reduction of total harmonic distortion (THD) of grid are achieved in this system. Its applicability and reliability are demonstrated by various simulated results using MATLAB/Simulink platform and hardware implementation.

Keywords—Power flow control; Solar photovoltaic; Brushless DC motor; Voltage source converter; Unit vector template; Voltage source inverter; Maximum power point; Power quality; Power factor; Total harmonic distortion.

I. INTRODUCTION

The continuously increasing carbon emission and diminishing of fossil fuels encourage the instant consumers to adopt the renewable energy. A solar photovoltaic (PV) generation is emerging as the best alternative of conventional sources for various appliances [1]. With reference to this, the water pumping has gained a broad attention in last few decades as a crucial application of PV energy [2-3]. The DC motors have been used initially to pump the water followed by an AC induction motor [4]. An innumerable researches have been carried out on electric motor drives to improve the performance and efficiency of PV fed pumping systems with cost benefit. A permanent magnet brushless DC (BLDC) motor, due to its high efficiency, high power density, no maintenance, long service life, low electromagnetic interference (EMI) issues and small size, is being opted from last decade [5]. It has been determined that introducing this motor reduces the cost and size of PV panels in addition to improved performance and maintenance free operation [6].

Being a grid-isolated or standalone system, the existing BLDC motor driven water pumps fed by a PV array rely only on solar PV energy. Due to its intermittency, the solar PV generation exhibits its major drawbacks, which results in an unreliable water pumping systems. In the course of bad climatic condition, water pumping is severely interrupted, and the system is underutilized as the pump is not operated at its full capacity. Moreover, an unavailability of sunlight (at night) leads to shutdown of the water pumping system. These shortcomings are required to be overcome in order to acquire a

reliable PV based pumping system. Few attempts in this connection are found in [7-10], although not with BLDC motor drive, which deploy a battery as an energy storage. Associated with a bidirectional control, the battery is charged and discharged during full and poor solar radiation (or no radiation) respectively, thus it ensures a full water delivery continuously. Contrary to it, introducing a battery energy storage in PV based water pumping not only increases the overall cost and maintenance but also reduces its service life [11-12]. A lead acid battery which is mostly used, has a useful life of only 2-3 years [13].

The aforementioned demerits with the battery storage have turned the attention towards an alternate technological solution which may be best suited in every aspect for a reliable water pumping based on PV generation. These recently recognized technologies, in reality, interface a PV generating unit which is installed for water pumping, into a utility grid. The prime attention is to achieve an uninterrupted water pumping with its full capacity regardless of operating conditions, whether day or night. A grid connected solar water pumping system is reported in [14] wherein a power allocation system decides whether to draw power from PV array or from the utility (when PV array is insufficient to power the pump). A water pump along with a pump controller is connected at the common DC bus of PV array and grid connected inverter. No battery storage is used, a service life of the system is thus prolonged, and the maintenance and manufacturing cost are reduced. However, the developed control enables only a unidirectional power flow i.e. an excess power or an unutilized power (when pumping is not required) of PV array is not returned to the utility grid. Therefore, the PV installation is not fully utilized and a consumer must pay an electricity bill.

Such another system [13] first feeds the PV energy into the utility grid through a grid inverter and a water pump is then fed by that utility grid through a pump inverter. Although being a grid connected PV pumping system, it appears as a system operated by utility grid only. A kind of hybrid PV water pumping is presented in [15], wherein a battery is first charged by PV array through a charge controller and then it is discharged to feed the water pump via an inverter. The pump is also supported by a utility interface through an option switch. This system becomes expensive due to an added manufacturing and maintenance cost of the battery storage. A part of the PV installation is engaged in water pumping and the remaining part in feeding power to the grid in [16-17]. The system is not reliable as the pumping is dependent only on the PV energy and no power is drawn from the utility. A grid interfaced PV fed-BLDC motor driven water pumping with unidirectional power flow control is developed in [18], wherein the remaining power is drawn from the grid whenever required. The developed system fails to utilize the PV power in case the water pumping is not required.

All these aforementioned existing topologies of a PV based pumping systems present a unidirectional power flow control which either feeds the grid or draws power from the grid. A multifunctional system which may enable a bi-directional power flow depending on the operating circumstances such that both PV installation and pumping system are fully

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utilized, is yet to be developed. This work presents suchlike system employing first-time a BLDC motor drive. As mentioned, the proposed system deals with the development of a bi-directional power flow control, enabling the flow of power from PV array to the single phase utility grid in case a water pumping is not required, and from the grid to BLDC motor-pump in case the PV array power is not sufficient (or at night) to run the pump at its full capacity. This practice offers a source of earning to the consumers by sale of electricity to the utility. A unit vector template (UVT) generation, due to its simplicity and ability to serve the objective, is applied to perform a bi-directional power transfer. The proposed system also meets the power quality standards required by a utility grid as per IEEE-519 standard [19]. A grid interfaced PV based water pumping system, incorporating some of the aforementioned features, has been reported in [20]. A detailed design approach, control methodology, simulation analysis and hardware implementation are added here.

The MPPT (Maximum Power Point Tracking) of PV array [21-25] is achieved by an incremental conductance (InC) technique using a DC-DC boost converter. The magnitude of stator current of BLDC motor at starting is controlled by operating the VSI (Voltage Source Inverter) in PWM (Pulse Width Modulation) mode for a pre-defined duration. However, once the motor is started, the VSI is operated with the pulses of fundamental frequency resulting in a minimized switching loss and an enhanced conversion efficiency. Moreover, no phase current sensor is used for BLDC motor control which leads to an increased cost benefit. The grid interactive PV based water pumping system using a BLDC motor drive is designed, modelled and simulated in MATLAB/Simulink platform, and its performances are evaluated through simulated results followed by hardware implementation to demonstrate the claims. The main contributions of this work are as follows:

- The BLDC motor-driven-solar water pumping system is interfaced to the utility grid in order to develop a reliable water pumping.
- A bidirectional power flow control is proposed, which leads to full volume of water delivery, continuously, regardless of the weather condition.
- The proposed system is controlled such that the power generated by the PV array is transferred to the utility grid in case the water pumping is not required. This feature offers a full utilization of the installed resources. Moreover, the system emerges as a source of income by sale of electricity to the utility.
- The proposed water pumping system is designed and controlled such that it also continues to deliver the water in case of grid failure. The volume of water delivery then depends on the available solar radiation.
- The three phase VSI, feeding the BLDC motor is switched at fundamental frequency. This results in a considerable reduction in the switching loss associated with the said VSI.
- Power quality standards at the grid are maintained as the IEEE-519 standard regardless of the direction of power flow (during both feeding and drawing of the power).

II. CONFIGURATION OF PROPOSED SYSTEM

A configuration of the proposed water pumping system is presented in Fig. 1, wherein a BLDC motor runs a water pump. A PV array feeds a BLDC motor-pump via a boost converter and VSI. The boost converter performs MPPT of PV array through InC algorithm while the VSI performs an electronic commutation of BLDC motor [5, 26]. An inbuilt

encoder generates three Hall-Effect signals to carry out an electronic commutation. The DC bus of VSI is supported by a single phase utility grid. A voltage source converter (VSC) enables a bi-directional power transfer through a DC bus capacitor. The PV array feeds the grid only when a water pumping is not required otherwise it is a preferred objective. An interfacing inductor is placed in the line to allow power flow between the grid and VSC, and to limit the harmonics current into the supply. A RC ripple filter is provided to limit the harmonics on supply voltage. An integrated mathematical modelling of the overall system is given in Appendices.

III. SPEED CONTROL OF BLDC MOTOR

As discussed before, the proposed BLDC motor drive eliminates the phase current sensors. It is desired to operate the BLDC motor-pump at its rated speed irrespective of the climatic condition. This is achieved by continuously regulating the DC bus voltage of VSI at the rated DC voltage of BLDC motor. A bi-directional power flow control enables, by regulating the DC bus voltage and hence the operating speed, to deliver a full amount of power required to pump the water with full capacity. In case the grid is not available, the DC bus voltage is not maintained at the rated DC voltage of BLDC motor under bad climatic conditions, and the speed is governed by a variable DC bus voltage.

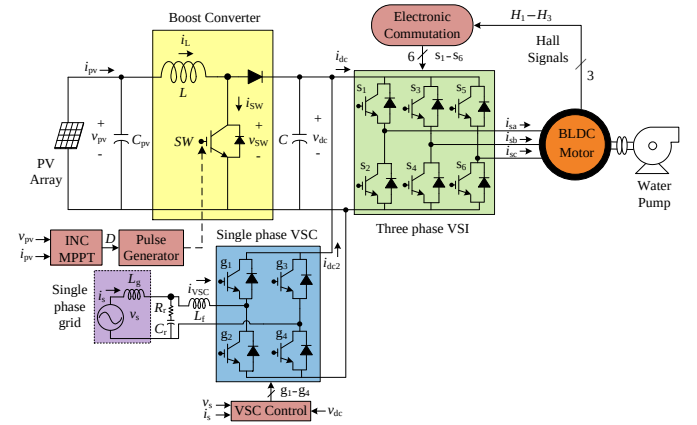


Fig.1 Schematic of the grid interactive PV array based water pumping system using a BLDC motor drive

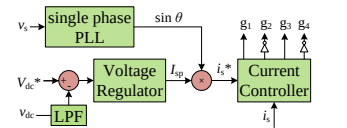


Fig. 2 UVT based bi-directional power flow control of VSC

IV. BI-DIRECTIONAL POWER FLOW CONTROL

The development of a reliable water pumping system and full utilization of the resources are realized by a grid interactive PV generation. To allow the flow of power in either direction, a bi-directional power control based on a UVT generation [20, 27-28] is applied as shown in Fig. 2. This is the simplest technique and is easy to implement as it does not require any complex mathematical model or algorithm. A single phase PLL (Phase Locked Loop) is used to synchronize the utility grid voltage and current. It generates a sinusoidal unit vector of supply voltage, $\sin \theta$ at fundamental frequency. On the other hand, an amplitude of fundamental component of supply current, I_{sp} is extracted by regulating the DC bus voltage, v_{dc} . A proportional-integral (PI) controller is used as a voltage regulator. v_{dc} is sensed and passed through a first-order low pass filter to suppress the ripple contents. The

filtered v_{dc} is then compared with a set value, V_{dc}^* . A fundamental component of supply current, i_s^* is extracted by multiplying I_{sp} and $\sin \theta$. The sensed supply current, i_s is compared with i_s^* and error is processed through a current controller to generate the gating pulses for VSC.

When it is required to draw power from utility, the voltage regulator generates a positive I_{sp} . Therefore, an in-phase supply current is drawn from the grid. Likewise, when the utility is fed by PV array, a negative I_{sp} is generated resulting in an out-of-phase supply current. Thus, by reversing the direction of current, direction of power flow is controlled as per the requirement. An improved power quality at the utility grid is also ensured by the applied control technique in terms of total harmonic distortion (THD) and power factor. In case the grid is not available, the DC bus voltage cannot be regulated. Nevertheless, the PV array is able to feed the water pump in standalone mode although being sensitive to the climatic condition. The detailed analysis of proposed bidirectional power flow control is given in Appendices.

V. SYSTEM DESIGN

Various operating stages associated with the configuration shown in Fig. 1 viz. PV array, boost converter, single phase grid, single phase VSC, three phase VSI and BLDC motor, are selected properly in order to get a good performance with their optimum design. The design approaches of these components are given in this section.

A. Design of PV Array

A 1.5 kWp-PV array is designed for a 1.3 kW-BLDC motor-pump. The power losses associated with the converters and the motor-pump are taken into account. The parameters are estimated at the standard test condition (1000 W/m², 25°C, AM 1.5). A PV module-BMU/214 with an MPP voltage of 28.5 V and an MPP current of 7.5 A (complete specifications are given in Appendices) [29] is chosen to design a PV array of required capacity. First, the voltage of PV array, at MPP, is selected in view of the DC voltage rating of BLDC motor i.e. the DC bus voltage of VSI. It is selected as $V_{mpp} = v_{pv} = 200$ V, and the remaining parameters are accordingly calculated as, The current at peak power point,

$$I_{mpp} = i_{pv} = \frac{p_{pv}}{v_{pv}} = 1500/200 = 7.5 \text{ A} \quad (1)$$

where $p_{pv} = P_{mpp} = 1500$ W is the power of PV array at MPP. Series modules are as,

$$N_s = \frac{V_{mpp}}{V_m} = 200/28.5 = 7 \quad (2)$$

Parallel modules are as,

$$N_p = \frac{I_{mpp}}{I_m} = 7.5/7.5 = 1 \quad (3)$$

where V_m and I_m are MPP-voltage and MPP-current of a module.

According to (2) and (3), seven modules are connected in series to design a PV array of required size.

B. Design of Boost Converter

The design of a boost converter consists of an estimation of input inductor, L . It is selected in a manner to operate the converter in CCM, irrespective of the weather condition. The duty ratio, D_1 is estimated as [30],

$$D_1 = \frac{V_{dc} - v_{pv}}{V_{dc}} = \frac{270 - 200}{270} = 0.25 \quad (4)$$

where $V_{dc} = 270$ V is the DC bus voltage of the VSI.

The inductor, L is estimated as,

$$L = \frac{D_1 v_{pv}}{f_{sw} \Delta I_L} = \frac{0.25 \times 200}{10000 \times (7.5 \times 0.2)} = 3.3 \text{ mH} \quad (5)$$

where f_{sw} is the switching frequency of boost converter; ΔI_L is ripple in the current through L , $I_L (= I_{mpp})$.

C. Design of Three Phase VSI

A three phase VSI is used to feed the BLDC motor. Its design consists of an estimation of voltage, current and VA ratings. As the DC bus voltage is 270 V, the required voltage rating of an IGBT switch is calculated as,

$$V_{VSI} = V_{dc} \times 1.4 = 270 \times 1.4 = 378 \approx 400 \text{ V} \quad (6)$$

A voltage safety factor of 1.4 is selected to accommodate the switching transients. Similarly, the required current rating of an IGBT switch is calculated as,

$$I_{VSI} = I_{dc} \times 1.3 = (1500/270) \times 1.3 = 7.23 \approx 7.5 \text{ A} \quad (7)$$

where 1.3 is a current safety factor.

Finally, the required VA rating of VSI is estimated as,

$$VA_{VSI} = V_{VSI} \times I_{VSI} = 400 \times 7.5 = 3 \text{ kVA} \quad (8)$$

D. Design of Single Phase VSC

A single-phase VSC is used to control a bidirectional power flow. In a single-phase VSC, the blocking voltage of switching devices is equal to the DC link voltage. As the DC link voltage is 270 V, the switches have to block this voltage. A safety factor of 1.4 is selected to accommodate the voltage transients due to a high frequency switching. Therefore, the voltage rating of the IGBT devices are estimated as,

$$V_{VSC} = V_{dc} \times 1.4 = 270 \times 1.4 = 378 \approx 400 \text{ V} \quad (9)$$

The VSC has a maximum current drawn from the grid or to be fed to the grid. The said current is estimated as,

$$I_{s,max} = \sqrt{2} \frac{P_{mpp}}{V_s} = \sqrt{2} \frac{1500}{180} = 11.78 \text{ A} \quad (10)$$

where $V_s = 180$ V is rms value of the utility grid voltage.

Thus, the maximum current rating of IGBT devices is 11.78 A. Considering a safety factor of 1.3, the current rating is estimated as,

$$I_{VSC} = I_{s,max} \times 1.3 = 11.78 \times 1.3 = 15.3 \approx 15 \text{ A} \quad (11)$$

Finally, the required VA rating of VSC is estimated as,

$$VA_{VSC} = V_{VSC} \times I_{VSC} = 400 \times 15 = 6 \text{ kVA} \quad (12)$$

E. Design of Common DC Link Capacitor

The DC link capacitor, C is common to the boost converter, three phase VSI and single phase VSC. It is tuned for the second harmonic component of single phase grid voltage, which is appeared on the DC bus of a single phase VSC. Thus, the capacitor, C is estimated as [31],

$$C = \frac{I_{dc}}{2\omega_L \Delta V_{dc}} = \frac{1500/270}{2 \times (2\pi \times 50) \times (270 \times 0.008)} = 4700 \text{ } \mu\text{F} \quad (13)$$

where I_{dc} is an average current flowing through the DC bus, ω_L is the line frequency in rad/s, and ΔV_{dc} is ripple in the DC bus voltage.

F. Design of Interfacing Inductor

The selection of an interfacing inductor, L_f depends on the permitted current ripple ΔI_{VSC} . It is estimated as,

$$L_f = \frac{mV_{dc}}{4af_{sw}\Delta I_{VSC}} = \frac{1 \times 270}{4 \times 1.2 \times 10000 \times (1500/180) \times 0.2} = 3.3 \text{ mH} \quad (14)$$

where modulation index, $m=1$, over loading factor, $a=1.2$, switching frequency, $f_{sw}=10$ kHz, current ripple, $\Delta I_{VSC}=20\%$ of I_{VSC} .

G. Design of R-C Ripple Filter

A first order high pass filter is used to suppress the switching harmonics generated by the VSC. A small R-C filter is connected at the utility grid side. This ripple filter is designed such that it offers very high impedance to the fundamental frequency component and low impedance to the switching frequency component. In order to meet this condition, $R_f C_r \ll T_{sw}$, where R_f , C_r and T_{sw} are respectively the ripple filter resistance, ripple filter capacitance and switching time.

Considering $R_f C_r = T_{sw}/4$, $T_{sw} = 1/10000$ s and $R_f = 5 \Omega$, C_r is estimated as [32],

$$C_r = \frac{T_{sw}}{4R_f} = \frac{1}{10000 \times 4 \times 5} = 5 \mu\text{F} \quad (15)$$

Thus, a series combination of 5Ω resistance and $5 \mu\text{F}$ capacitance is selected as a RC ripple filter.

VI. SIMULATION RESULTS AND DISCUSSION

An analysis of the proposed system under various operating conditions is carried out through the simulated results in MATLAB/Simulink platform. The developed system and its control are tested for starting, dynamic, and steady state operations. A 4-pole, 3000 rpm @ 270 V (DC), 1.3 kW motor-pump is powered by a 1.5 kWp (under standard test conditions) PV array and a single phase 180 V, 50 Hz utility grid. Detailed specifications of the system are given in Appendices. The water pump is operated with a PV array only, with the grid only, with both PV array and grid, or may not be operated for instance. All these possible operating conditions are considered for the demonstration of proposed system.

A. Starting and Steady State Performance

The main objectives of these performance studies are to demonstrate the soft starting of BLDC motor and steady state operation of motor-pump under various operating conditions.

1) *When Only PV Array Feeds BLDC Motor-Pump:* Various PV array and BLDC motor-pump indices are presented in Fig. 3. As shown in Fig. 3(a), PV array is operated at its MPP under the radiation level of 1000 W/m^2 . Therefore, the BLDC motor-pump is also operated at its full capacity and it runs at rated speed i.e. 3000 rpm, as shown in Fig. 3(b). No grid power is required as the PV array generates a sufficient power to run the pump at its full capacity. The various indices refer to back-EMF, e_a , stator current, i_{sa} , speed, N , electromagnetic torque, T_e , and load torque, T_L . These results demonstrate a soft starting along with the successful steady state operation of the motor-pump.

2) *When Only Utility Grid Feeds BLDC Motor-Pump:* This operating condition occurs when a water pumping is required at night. Fig. 4(a) depicts that an in-phase sinusoidal

supply current of 8.3 A (rms) is drawn and DC bus voltage is maintained at 270 V. The motor draws a sufficient power from utility to run at full capacity, as shown in Fig. 4(b). A full utilization of pumping system is demonstrated in this case.

3) *When Water Pumping is not required:* In this case, the pump is not operated and power generated by the PV array is fed to the utility grid. Fig. 5(a) shows the MPP operation of PV array at 1000 W/m^2 . Fig. 5(b) exhibits an out-of-phase sinusoidal supply current which indicates that the utility is fed by a PV array and the power flow is reversed while maintaining the DC voltage at 270 V.

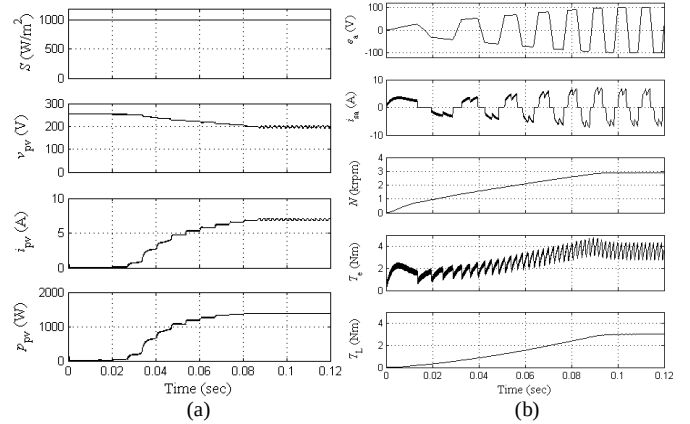


Fig. 3 Steady state and starting performance of (a) PV array, and (b) BLDC motor-pump, when only PV array feeds BLDC motor-pump

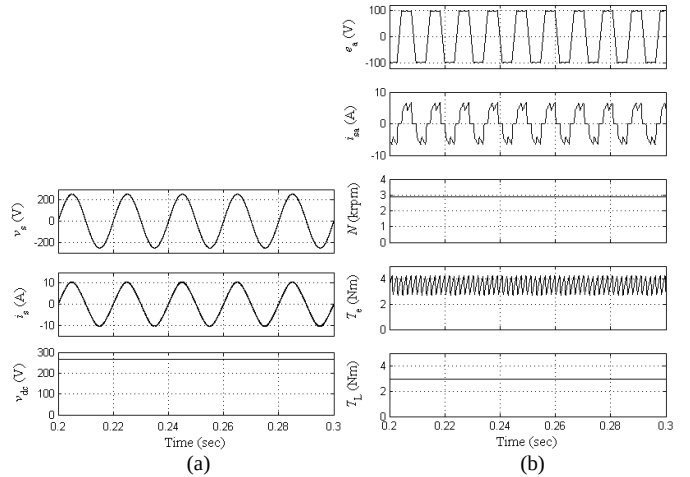


Fig. 4 Starting and steady state performance of (a) utility grid, and (b) BLDC motor-pump, when only utility grid feeds water pump

B. Dynamic Performance

A sudden variation in climatic condition or requirement of sudden change in the direction of power flow, is considered as a dynamic condition. The proposed system is tested to operate successfully under the considered dynamics.

1) *Transition from Grid Feeding Pump to PV Array Feeding Grid:* This analysis assumes that the water pump is operated initially through utility grid when PV array power is not available. The mode of operation is suddenly changed by considering that the water pumping is no more required but PV array power is available. Therefore, it is desired now to feed the utility by PV array. Fig. 6 illustrates the performance under this dynamic condition. The mode of operation is changed at 0.3 s. Figs. 6(a), 6(b) and 6(c) respectively present the PV array indices, utility grid indices and BLDC motor-pump indices. As shown in Fig. 6(b), the direction of current flow reverses within a half cycle. The DC bus voltage is regulated at 270 V as shown in Fig. 6(c). A reduction in the

speed starts at 0.3 s and motor-pump reaches standstill after a while as shown in Fig. 6(c).

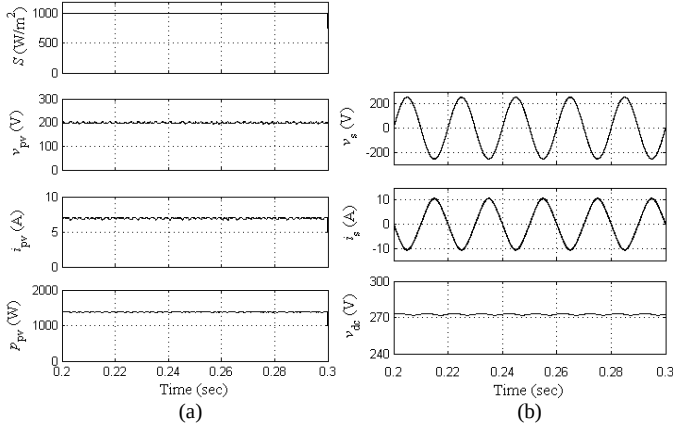


Fig. 5 Steady state response of (a) PV array and (b) utility grid, when water pumping is not required

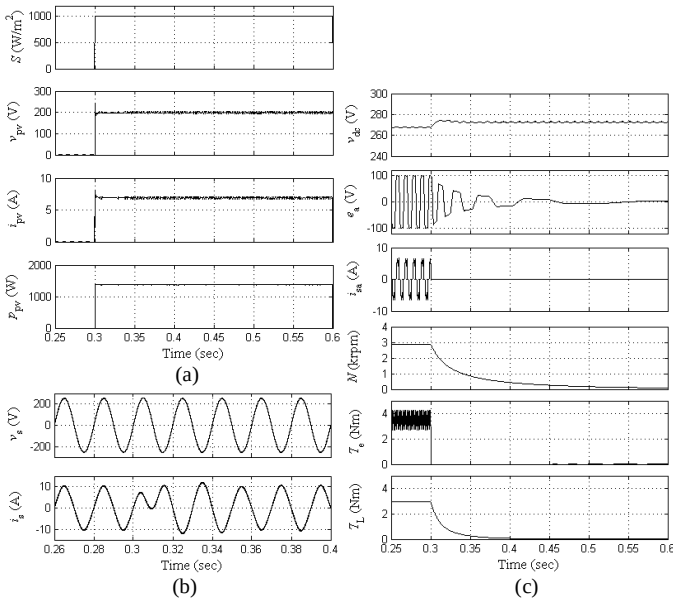


Fig. 6 Dynamic response of (a) PV array (b) utility grid and (c) BLDC motor, under a transition from grid feeding pump to PV array feeding grid

2) Transition from PV Array Feeding Pump to Both PV Array and Grid Feeding Pump: In this case it is assumed that only the PV array is feeding the pump initially as it is sufficient to run the water pump at its full capacity. A reduction in radiation level from 1000 W/m^2 to 500 W/m^2 is observed at 0.3 s. Since the PV array alone is unable to run the water pump at its full capacity at 500 W/m^2 , it is desired to draw the remaining power from the utility. This condition is illustrated in Fig. 7. As shown in Fig. 7(a), the maximum PV array power reduces corresponding to a radiation level of 500 W/m^2 . Prior to 0.3 s, no power is drawn from the utility as shown by i_s in Fig. 7(b). From 0.3 s onwards, the remaining power is drawn from the utility resulting in a flow of in-phase supply current of 4.3 A (rms). Fig. 7(c) depicts that the motor-pump is operated at its full capacity regardless of climatic conditions. The motor runs at 3000 rpm as the DC bus voltage is regulated at 270 V.

C. Power Quality Aspects

The proposed system ensures an improved power quality on the utility grid in terms of power factor and THD. Fig. 8(a) presents the THD and harmonics spectrum of supply current, i_s shown in Fig. 4(a), in case the water pump is fed by utility grid only. Similarly, Fig. 8(b) presents the THD and

harmonics spectrum of i_s shown in Fig. 7(b), in case the radiation level is 600 W/m^2 and remaining power is required to be fed by the utility grid. Under both condition, THD of supply current is observed below 5% which meets the IEEE-519 standard. Moreover, a unity power factor operation is ensured under the various operating conditions.

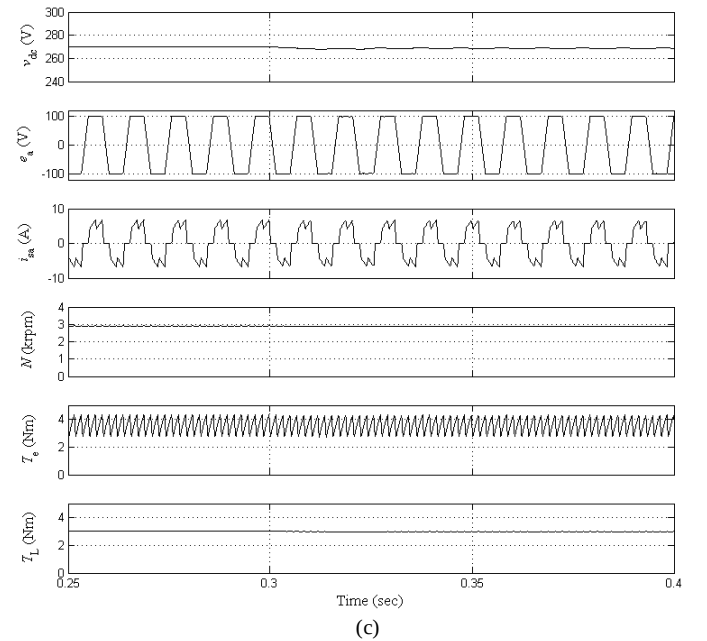
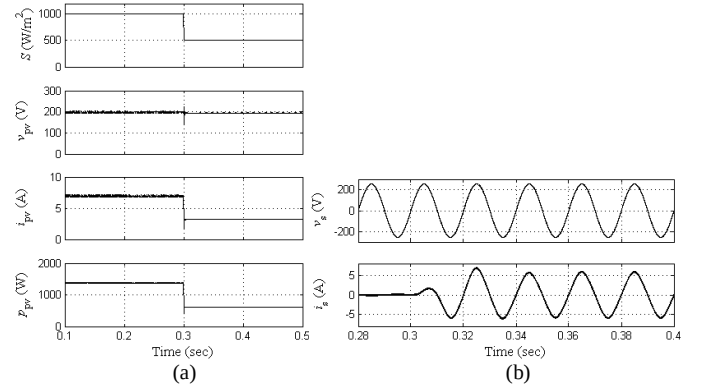


Fig. 7 Dynamic performance of (a) PV array (b) utility grid and (c) BLDC motor-pump, under a transition from PV array feeding pump to both PV array and grid feeding pump.

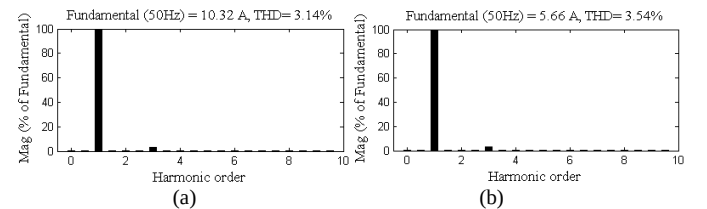


Fig. 8 THD and harmonic spectrum of supply current when the water pump is fed by (a) utility grid only, and (b) both utility grid and PV array.

VII. EXPERIMENTAL VALIDATION

A hardware prototype is developed to demonstrate the performances of the system. It consists of a solar array simulator (AMETEK make), a boost converter, a VSC (SEMIKRON make), a DSP-dSPACE 1104 controller, a BLDC motor (Motor Power Company make), and a DC generator (Benn make). The water pump is realized by a BLDC motor driven generator which feeds a resistive load. The power is proportional to square of the speed in this load. Both centrifugal pump and aforesaid loading arrangement experience a variation in power with respect to the speed. Thus, the system analysis and validation with the used loading

arrangement also indicate a successful operation with the centrifugal pump load. Tests are conducted with a 4 pole, 1.3 kW, 3000 rpm @ 270 V (DC) BLDC motor which is supplied by a solar array simulator, set at 1500 Wp, and supported by a single phase, 180 V, 50 Hz grid. A single phase power quality analyzer (Fluke make 43-B) is used to record power quality indices. The parameters of PV array, motor and single phase grid, are given in Appendices. Experimental performances of proposed topology are analyzed in following sections.

A. BLDC Motor-Pump is Fed by PV Array Only

When a full radiation is available, the required power is drawn from the PV array only, as shown in Fig. 9 (a). As the array feeds 1.5 kWp, the motor is run at its full speed and load. No power from the grid is required under this condition. Fig. 9 (b) shows recorded PV characteristics (power vs. voltage and current vs. voltage) at 1 kW/m², which indicates the MPP. A good tracking efficiency is observed as 99.7%.

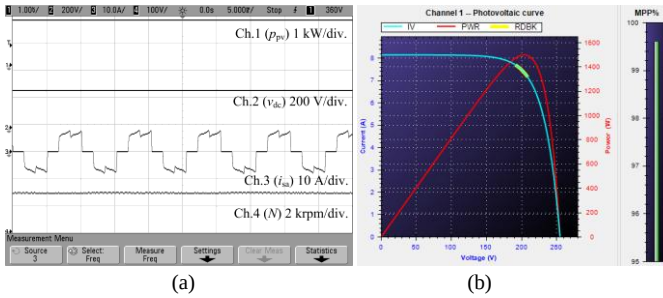


Fig. 9 Steady state responses when the motor-pump is fed by PV array only (a) PV array and BLDC motor-pump indices (b) PV array characteristics

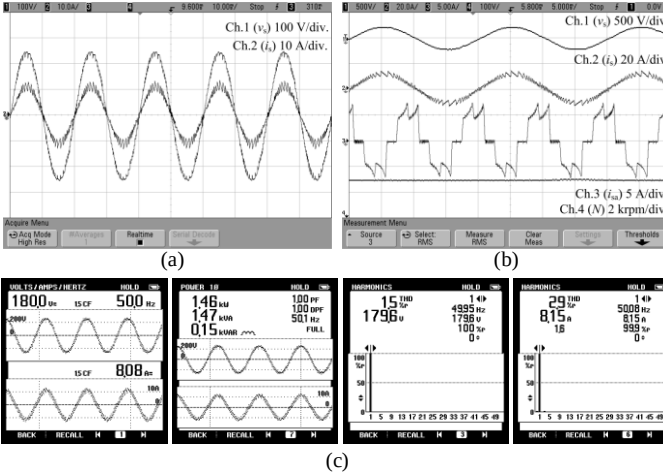


Fig. 10 Steady state performance, when the motor is fed by grid only (a) grid indices (b) grid and motor-pump indices (c) power quality at utility grid

B. BLDC Motor-Pump is Fed by Grid Only

Whenever the water pumping is required at night (in other words, when the solar power is not available), the required full power is drawn from the utility grid. This case is demonstrated in Fig. 10. An in-phase grid current, $i_s = 8.15$ A with respect to the grid voltage v_s , is drawn, as shown in Fig. 10 (a). The motor uninterruptedly runs at its rated speed and current as shown in Fig. 10 (b), which leads to a full water delivery. The improved power quality at the utility is certified by the waveforms and harmonic spectra shown in Fig. 10 (c). The supply voltage and current, power drawn, power factor, harmonic spectra of supply voltage and current, are shown. The current is drawn at unity power factor, and its THD is 2.9%. Moreover, the THD of supply voltage is measured as 1.5%. Therefore, the recorded waveforms certify a power quality improvement as per the IEEE-519 standard.

C. Grid is Fed by PV Array

As already discussed, the power generated by a PV array is fed to the utility grid in case the water pumping is not required. The steady state performance under this case is shown in Figs. 11-12. For instance, it is assumed that a full radiation of 1000 W/m² is available. The PV array indices under this radiation are shown in Fig. 11 (a), which depicts that the PV array generates 1.5 kWp. This power is fed to the grid as shown in Fig. 11 (b). The supply current is out of phase with respect to the supply voltage, and the DC bus voltage is maintained at 270 V. A zoomed view of supply voltage and current is also presented in Fig. 11 (c). The power quality indices are also presented for this case as shown in Fig. 11 (d). These indices confirm the improved power quality at ac mains as per the IEEE-519 standard.

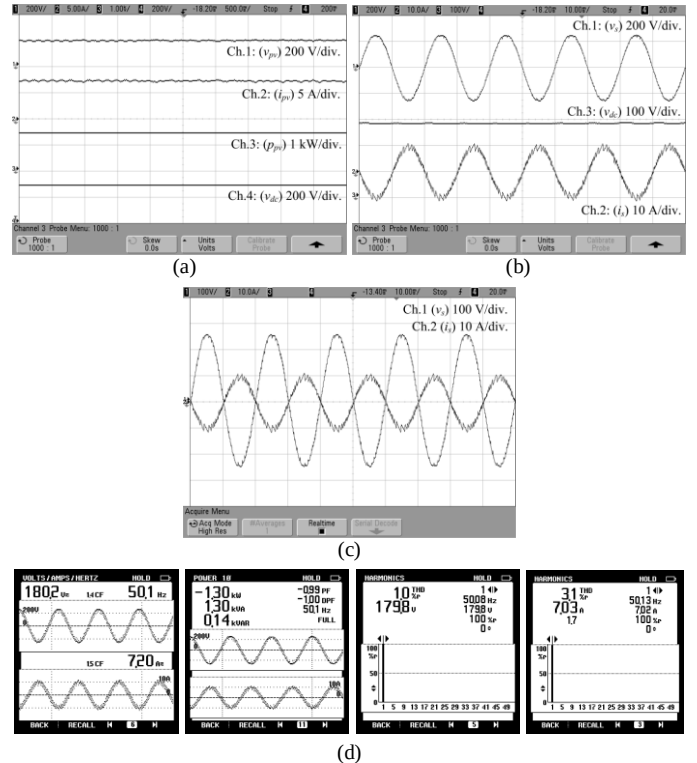


Fig. 11 Steady state performance of (a) PV array (b)-(c) utility grid, and (d) power quality at utility grid, when PV array feeds grid at 1000 W/m²

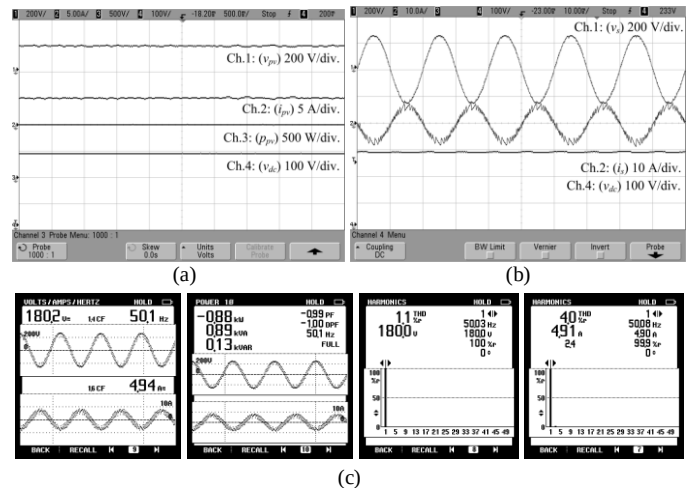


Fig. 12 Steady state performance of (a) PV array (b) utility grid, and (c) power quality at utility grid, when PV array feeds grid at 400 W/m²

As the radiation level drops, the power generated by solar PV array also reduces. However, whatsoever power is generated corresponding to that particular radiation level, is

successfully fed to the grid. This case is expressed in Fig. 12. As presented in Fig. 12 (a), a 1050 Wp is generated by the PV array corresponding to a radiation level of 700 W/m². This amount of power is fed to the grid as shown in Fig. 12 (b). The DC bus voltage is consistently regulated at 270 V and the power is fed at unity power factor. The various grid indices are also shown in Fig. 12 (c), to ensure the improved power quality at single phase grid.

The proposed system should properly respond to any variation in the weather profile, in order to manifest its workability under dynamic conditions. The system responses under dynamically varying radiation level are shown in Figs. 13-14. A step change in the radiation level from 1000 W/m² to 700 W/m² is observed, and the response of PV array to this variation is plotted in Fig. 13 (a). The generated power reduces corresponding to 700 W/m². Therefore, a power fed to the grid instantly reduces, as depicted by the grid current, i_s in Fig. 13 (b). The power flow control regulates the DC bus voltage at 270 V. A zoomed view of v_s and i_s on a common axis is also presented in Fig. 13 (c). An opposite case is represented in Fig. 14, wherein the PV array is gone through a sudden increment in solar radiation, from 700 W/m² to 1000 W/m². The system responds satisfactorily to this instant variation. The power generated is indeed increased as shown in Fig. 14 (a). The power fed to the utility is thereby increased as observed by i_s in Fig. 14 (b). The DC bus voltage is maintained regardless of the variation in power generated.

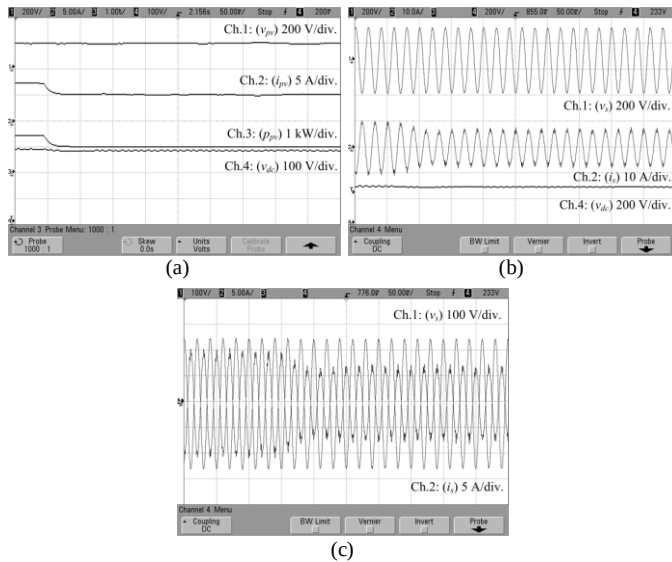


Fig. 13 Dynamic performance of (a) PV array and (b)-(c) grid, when PV array feeds grid under a variation in radiation level from 1000 W/m² to 700 W/m²

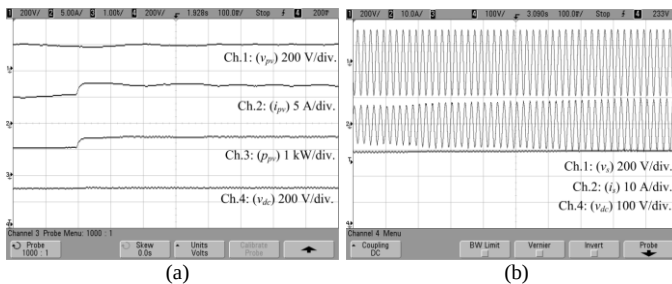


Fig. 14 Dynamic performance of (a) PV array and (b) grid, when PV array feeds grid under a variation in radiation level from 700 W/m² to 1000 W/m²

D. BLDC Motor-Pump is Fed by Both PV Array and Grid

Under a bad weather condition, the PV array generated power becomes insufficient to run a water pump at its full capacity. That is exactly the case where a utility grid support is

required. The proposed power flow control estimates an amount of power required to be drawn from the grid such that a full power is transferred to the motor pump. To demonstrate this case, it is considered that a radiation of 400 W/m² is available sometime. A steady state response with this radiation is shown in Fig. 15 (a). Merely a 605 Wp is generated by the PV array. Nevertheless, the motor attains a rated speed and it draws a rated current. This is made possible by drawing a requisite amount of power from the grid. A 4.85 A (rms) is drawn from the utility. Ultimately the pump delivers a full volume of water. Fig. 15 (b) consists of the grid indices and power quality information such as THD and power factor. THDs of grid voltage and current are within the limit of the IEEE-519 standard. Fig. 15 (b) also depicts the required power drawn (0.89 kVA) and an rms current drawn (4.85 A).

The behavior of the system under a dynamic variation in climatic condition is also tested as shown in Fig. 16. The solar radiation is reduced from 1 kW/m² to 400 W/m². As a result, the power generated by PV array, p_{pv} reduces as shown in Fig. 16 (a). As the radiation drops, in order to ensure a full volume of water delivery, the required current, i_s is drawn from the utility grid. This enables the motor-pump to run at its rated speed, $N = 3000$ rpm, while regulating the DC bus voltage at 270 V. No current is being drawn from the grid before a drop in the radiation. Conversely, a case is considered wherein a radiation abruptly increases from 400 W/m² to 1000 W/m². The power generated by PV array reaches 1.5 kWp, as shown in Fig. 16 (b), which is sufficient to deliver a full amount of water. As the MPP is reached, the grid current becomes zero. This denotes that the power is no longer drawn from the grid. The motor continues to run at its rated speed as the rated DC bus voltage is maintained. It is clearly visible that a requisite current is drawn from the grid before increasing the radiation.

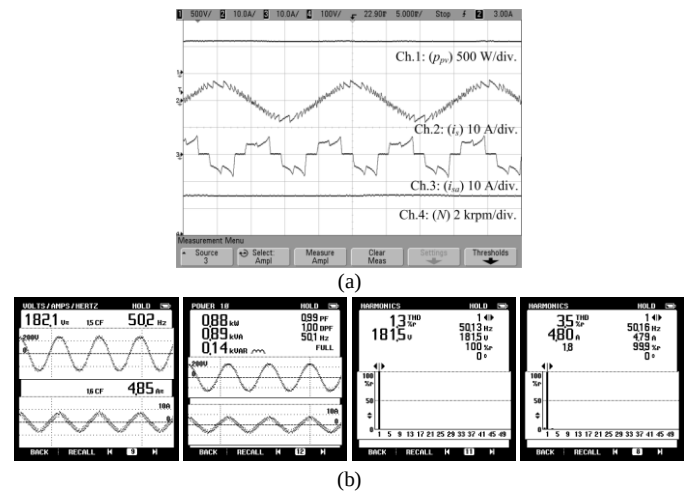


Fig. 15 Steady state performance, when motor-pump is fed by both PV array and grid (a) PV array, grid and motor indices (b) power quality indices of grid

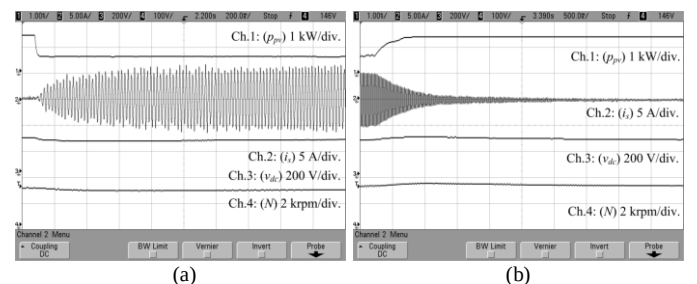


Fig. 16 Dynamic performance, when BLDC motor-pump is fed by both PV array and grid (a) radiation reduces from 1000 W/m² to 400 W/m² (b) radiation increases from 400 W/m² to 1000 W/m²

E. Transition from PV Array and Grid Feeding Pump to PV Array Feeding Grid

This transition is possible if the motor-pump is run with a partial support of the utility grid and the water pumping is no more required. The power available from the PV array is desired to be fed to the utility grid. Fig. 17 illustrates this considered dynamic condition. Prior to the transition, the PV array is operated at 400 W/m². Therefore, the motor-pump is fed with the support of utility grid. As the motor-pump system is disconnected from the DC bus, the direction of power flow reverses and the grid current appears out-of-phase with the grid voltage within a cycle, as shown in Fig. 17 (a). The DC bus voltage, v_{dc} is regulated at its rated value, as presented in Fig. 17 (b). The stator current becomes zero as the motor-pump is disconnected. These results demonstrate a successful transition under the considered dynamic condition.

F. Grid Failure Condition

In case the utility grid fails during an operating mode wherein the grid feeds a fraction of power to the motor-pump, the proposed grid interactive solar PV fed water pumping system becomes a standalone PV-water pumping system. Nevertheless, the system continues to perform successfully, although the motor-pump no longer runs at its full capacity. Fig. 18 incorporates a sudden grid failure, and presents the response of the system under said dynamics. An instant before the grid failure, the pumping system is run at 700 W/m². It is operated at its rated condition with a utility grid support. As the grid collapses and an amount of power drawn from the grid becomes null, the motor speed and current are reduced corresponding to 700 W/m². However, the system still works in standalone mode, and it enables a water delivery corresponding to an available PV array power at 700 W/m².

The obtained experimental results are precisely similar to that obtained from the simulation. Thus, the veracity of the simulation model is established.

An amount of power flow from PV array and utility grid to motor-pump or from PV array to the utility grid depends on various operating conditions. Table I shows the power distribution at different solar radiations in order to accomplish the desired function.

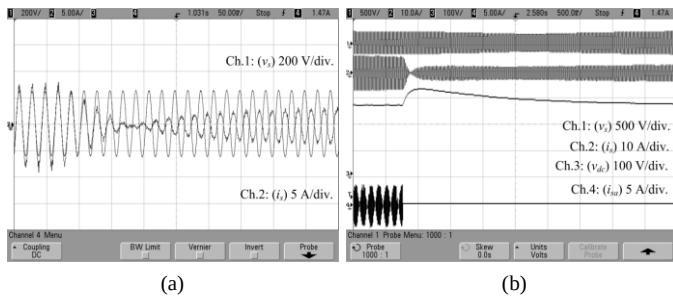


Fig. 17 Dynamic performance under transition from PV array and grid feeding pump to PV array feeding grid (a) grid indices (b) grid and motor indices

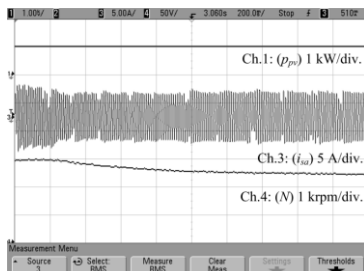


Fig. 18 Dynamic performance under grid failure

TABLE I
POWER DISTRIBUTION UNDER VARIOUS OPERATING CONDITIONS

Solar radiation (W/m ²)	Water pumping is required		Water pumping is not required
	Power drawn from PV array (W)	Power drawn from grid (W)	Power fed to grid (W)
1000	1500	0	1300
900	1348	150	1167
800	1195	301	1030
700	1046	444	890
600	897	592	765
500	747	740	634
400	605	890	501
300	449	1040	365
200	294	1188	234
100	144	1335	105
0	0	1470	0

VIII. CONCLUSIONS

A single phase grid interactive PV array based water pumping system using a BLDC motor drive has been proposed and demonstrated. A bi-directional power flow control of VSC has enabled a full utilization of resources and water pumping with maximum capacity regardless of the climatic conditions. A simple UVT generation technique has been applied to control the power flow as desired. All the power quality aspects have been met as per the IEEE-519 standard. The speed control of BLDC motor-pump has been achieved without any current sensing elements. A fundamental frequency switching of VSI has contributed to enhance the efficiency of overall system by reducing the switching losses. The proposed solution has emerged as a reliable water pumping system, and as a source of earning by sale of electricity to the utility when water pumping is not required.

APPENDIX A

Parameters of Solar Array

Peak power = 1.5 kW; Open circuit voltage = 254.8 V; MPP voltage = 200 V; Short circuit current = 8.15 A; MPP current = 7.5 A.

Specifications of Motor

Poles = 4; Speed = 3000 rpm; Stator resistance = 3.58 Ω ; Stator inductance = 9.13 mH; Voltage constant = 68 V/krpm.

Grid Interfacing Components

Interfacing inductor = 3.3 mH; R-C filter = 5 Ω , 5 μ F; DC bus capacitor = 4700 μ F.

APPENDIX B

MATHEMATICAL MODELLING OF PROPOSED SYSTEM

An integrated mathematical model of proposed system is developed in this section. The various system equations are derived for five stages: PV array, boost converter, single phase VSC, BLDC motor and water pump as follows.

PV Array Modelling: The averaged equation of PV array current, i_{pv} is given as [33],

$$i_{pv} = N_p I_{sc} - \frac{N_p I_{sc}}{\exp\left(\frac{V_{oc}}{AV_t}\right) - 1} \left[\exp\left(\frac{v_{pv}}{N_s AV_t}\right) - 1 \right] \quad (16)$$

where V_{oc} and I_{sc} are the open circuit voltage and short circuit current of PV array under standard atmospheric condition (1000 W/m², 25°C, AM 1.5), respectively. N_p and N_s are the number of modules connected in parallel and series, respectively. $V_t = kT/q$ is the thermal voltage of PV array with k is the Boltzmann constant ($1.38064852 \times 10^{-23}$), q is the

charge of electron ($1.60217662 \times 10^{-19}$) and T is the temperature (Kelvin) of the PV cells, and A is the diode ideality factor.

The linearised model is developed by the line tangent to the I - V curve at the MPP as,

$$i_{pv} = -K_{pv} V_{pv} \quad (17)$$

where

$$K_{pv} = \frac{N_p I_{sc}}{\left[\exp\left(\frac{V_{oc}}{A V_t}\right) - 1 \right] A N_s V_t} \exp\left(\frac{V_{mpp}}{A N_s V_t}\right) \quad (18)$$

This linearisation approach is also applicable for any other operating points.

A small capacitor, C_{pv} is used to suppress the ripple present in the PV array voltage as shown in Fig. 1. The dynamics of this capacitor is derived as,

$$C_{pv} \frac{dv_{pv}}{dt} = i_{pv} - i_L \quad (19)$$

DC-DC Boost Converter Modelling: The average switching period based approach is used here to derive the equations. The average inductor current is governed by [33-34],

$$L \frac{di_L}{dt} = v_{pv} - (1 - D_1) v_{dc} \quad (20)$$

Similarly, the average capacitor voltage is given as,

$$C \frac{dv_{dc}}{dt} = i_L (1 - D_1) + i_{dc2} - i_{dc} \quad (21)$$

where i_{dc2} is the current at the DC side of single phase VSC as indicated in Fig. 1.

Single Phase VSC Modelling: According to the working principle of a full bridge DC-AC converter, following averaged expressions are derived.

$$i_{dc2} = (2D_2 - 1) i_{VSC} \quad (22)$$

where D_2 is the duty cycle of the devices of VSC.

The current through the interfacing inductor, i_{VSC} is derived as,

$$L_f \frac{di_{VSC}}{dt} = (2D_2 - 1) v_{dc} - v_s \quad (23)$$

BLDC Motor Modelling: A dynamic model of the BLDC motor is developed in terms of time derivative of current, speed, and position. The per-phase voltages (v_a , v_b , and v_c) of the BLDC motor are given as [35-36],

$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = \begin{bmatrix} R_s & 0 & 0 \\ 0 & R_s & 0 \\ 0 & 0 & R_s \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} L & M & M \\ M & L & M \\ M & M & L \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix} \quad (24)$$

where i_a , i_b , and i_c are the phase currents, e_a , e_b , and e_c are the back EMFs, R_s is the phase resistance, L is the self-inductance, and M is the mutual inductance of the stator's winding of the BLDC motor.

The sum of the currents in three phases is zero for a three phase star-connected BLDC motor given as,

$$i_a + i_b + i_c = 0 \quad (25)$$

This leads to the simplification of the inductances matrix in the models as,

$$M i_b + M i_c = -M i_a \quad (26)$$

Substituting (26) in (24) gives,

$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = \begin{bmatrix} R_s & 0 & 0 \\ 0 & R_s & 0 \\ 0 & 0 & R_s \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} L-M & 0 & 0 \\ 0 & L-M & 0 \\ 0 & 0 & L-M \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix} \quad (27)$$

Now, rearranging (27), the current derivatives are obtained as,

$$\frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} = \begin{bmatrix} 1/(L-M) & 0 & 0 \\ 0 & 1/(L-M) & 0 \\ 0 & 0 & 1/(L-M) \end{bmatrix} \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} - \begin{bmatrix} R_s & 0 & 0 \\ 0 & R_s & 0 \\ 0 & 0 & R_s \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} - \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix} \quad (28)$$

and the electromagnetic torque is given as,

$$T_e = \frac{e_a i_a + e_b i_b + e_c i_c}{\omega_r} \quad (29)$$

The back EMF is also defined as,

$$e_x = f_x(\theta) \lambda_x \omega_r \quad (30)$$

where λ_x represents the flux, the functions $f_x(\theta)$ have the same shape as that of the back EMF, and “x” represents the phase “a,” “b,” or “c.”

Now, substituting (30) into (29)

$$T_e = \lambda_a f_a(\theta) i_a + \lambda_b f_b(\theta) i_b + \lambda_c f_c(\theta) i_c \quad (31)$$

The equation of motion is derived as,

$$\frac{d\omega_r}{dt} = \frac{T_e - T_m - B\omega_r}{J} \quad (32)$$

where T_m is the load torque, J is the moment of inertia of the motor, and B is the frictional constant.

The position derivative is expressed as,

$$\frac{d\theta_r}{dt} = \omega_r \quad (33)$$

Water Pump Modelling: The water pump is designed and modelled to run at its rated capacity under the standard atmospheric condition such that a full water delivery is achieved. The centrifugal pump possesses a torque-speed characteristics as [37],

$$T_m = c_1 \omega^2 + \text{sign}(\omega) \cdot (c_2 e^{-c_3 |\omega|} + c_4) \quad (34)$$

The first term in (34) is deduced from the affinity laws, given by (35). The constants k_1 , k_2 , k_3 depend on the settings of discharge valve. Generally, at high speed, the lower order terms are negligible.

$$P_m = k_1 \omega^3 - k_2 \omega^2 + k_3 \omega \approx k_1 \omega^3 \quad (35)$$

The remaining terms in (34) stand for a breakaway torque, needed to get over the static friction to start the pump. These

second and third terms refer the transition from static to kinetic friction and Coulomb friction, respectively. The impact of these frictions are neglected as the motor-pump attains 10-20% of its rated speed. Therefore, the pump is designed applying the pump affinity laws which possess a square torque-speed or a cubic power-speed characteristics.

Integrated Mathematical Modelling of Overall System: The modelling equations derived for each stage are combined here to develop an integrated modelling of the proposed system. Substituting i_{pv} from (17) in (19) gives,

$$C_{pv} \frac{dv_{pv}}{dt} = -K_{pv} v_{pv} - i_L \quad (36)$$

Substituting i_{dc2} from (22) in (21) gives,

$$C \frac{dv_{dc}}{dt} = i_L (1 - D_1) + (2D_2 - 1) i_{vsc} - i_{dc} \quad (37)$$

The DC bus current, i_{dc} in (37) can be expressed in terms of torque developed, T_e and torque constant, K_T as,

$$i_{dc} = \frac{T_e}{K_T} = \frac{\lambda_a f_a(\theta) i_a + \lambda_b f_b(\theta) i_b + \lambda_c f_c(\theta) i_c}{K_T} \quad (38)$$

Substituting i_{dc} from (38) in (37) gives,

$$C \frac{dv_{dc}}{dt} = i_L (1 - D_1) + (2D_2 - 1) i_{vsc} - \frac{\lambda_a f_a(\theta) i_a + \lambda_b f_b(\theta) i_b + \lambda_c f_c(\theta) i_c}{K_T} \quad (39)$$

Thereafter, the integrated model of the overall system is formed by (36) and (39), together with (20), (23), (28) and (32)-(33).

APPENDIX C

ANALYSIS OF BIDIRECTIONAL POWER FLOW CONTROL

As depicted in Fig. 2, the DC bus voltage regulation and grid synchronisation are achieved using current controlled VSC that inhibits two control loops; the outer DC-link voltage control loop and the inner grid current control loop. Fig. 19 shows the detailed block diagram of proposed power flow control with plant transfer functions. The main objective is to implement stable and dynamic current and voltage compensators (G_{i_PID} , G_{v_PID}). Hence, the models for both loops are required to plot their Bode plots in the frequency domain, and test for stability with adequate bandwidth and phase margin.

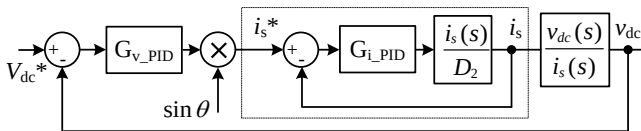


Fig. 19 Block diagram of inner current loop and outer voltage loop

The control to an input-current transfer function is obtained as [38],

$$\frac{i_s(s)}{D_2} = \frac{V_{dc} Z_r}{Z_g Z_r + Z_f (Z_g + Z_r)} \quad (40)$$

$$\text{where } Z_f = r_f + L_f s \quad (41)$$

$$Z_g = r_g + L_g s \quad (42)$$

$$Z_r = r_r + 1/C_r s \quad (43)$$

The r_f and r_g are the internal resistances of L_f and L_g , respectively.

The designed values of the aforesaid parameters are given as, $r_f = 0.05 \Omega$, $L_f = 3.3 \text{ mH}$, $r_g = 0.02 \Omega$, $L_g = 0.5 \text{ mH}$, $r_r = 5 \Omega$ and $C_r = 5 \mu\text{F}$. Therefore, the transfer function given in (40) is written as,

$$\frac{i_s(s)}{D_2} = \frac{1.688 \times 10^{-13} s^3 + 6.75 \times 10^{-9} s^2}{2.063 \times 10^{-22} s^5 + 2.386 \times 10^{-18} s^4 + 9.504 \times 10^{-14} s^3 + 1.75 \times 10^{-12} s^2} \quad (44)$$

The controller used for the control loop compensation is a well-known PID (Proportional-Integral-Derivative) controller described as,

$$G_{i_PID} = K_{ip} + \frac{K_{ii}}{s} + K_{id} s \quad (45)$$

Thus, the current loop gain is derived as,

$$G_{i_PID} \cdot \frac{i_s(s)}{D_2} = \left(K_{ip} + \frac{K_{ii}}{s} + K_{id} s \right) \cdot \frac{i_s(s)}{D_2} \quad (46)$$

The performance of the control loop is defined by the cross-over frequency and the phase margin. As the cross-over frequency directly impacts the THD, it must be as high as possible. On the other hand, the cross-over frequency must not be higher than 10-20% of the control-loop sampling frequency. The phase margin impacts the overshoot during the step change at the input of the PID controller.

A desired loop response with adequate crossover frequency and phase margin is obtained with $K_{ip} = 0.122$ and $K_{id} = 1.342 \times 10^{-6}$.

Fig. 20 shows the Bode plot of compensated current loop with a cross over frequency = 2.02 kHz and phase margin = 89.5°. It is also demonstrated that the system is stable.

The open loop gain for the outer voltage-control loop is derived by the energy balance equation, which is obtained as,

$$E_{dc} = E_g + \frac{1}{2} C v_{dc}^2 \quad (47)$$

where E_{dc} is the input energy to the DC-link, E_g is the energy captured by grid and $(1/2) C v_{dc}^2$ is the energy stored in DC link capacitor.

The relationship between variations in fundamental grid current and the DC-link voltage can be calculated using the average power balance equation derived from differentiation of (47) by time, assuming that the converter is lossless.

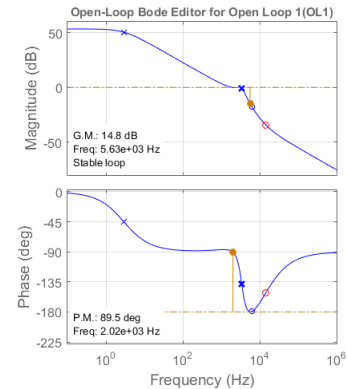


Fig. 20 Bode plot of compensated current loop

$$P_{dc} = P_g + \frac{d\left(\frac{1}{2}Cv_{dc}^2\right)}{dt} \quad (48)$$

For simplified sensitivity analysis, when studying relationship and correlation between certain system variables, other variables of least contribution and effect, on the studied variables, can be partially eliminated. Hence, for determining the impact of the grid current magnitude variation on the average DC-link voltage, P_{dc} is neglected [39]. Assume zero PV power, then $P_{dc}=0$, DC-link capacitor energy is solely affected by grid power as follows.

$$\frac{d\left(\frac{1}{2}Cv_{dc}^2\right)}{dt} = -P_g \quad (49)$$

$$\frac{d\left(\frac{1}{2}Cv_{dc}^2\right)}{dt} = -\frac{\hat{v}_s \hat{i}_s}{2} \quad (50)$$

Applying small perturbations around operating point leads to:

$$\frac{d\left[\frac{1}{2}C(v_{dc} + \Delta v_{dc})^2\right]}{dt} = -\frac{\hat{v}_s (\hat{i}_s + \Delta \hat{i}_s)}{2} \quad (51)$$

Neglecting steady-state values and square of small perturbations gives,

$$\frac{d\left[\frac{1}{2}C \times 2v_{dc} \Delta v_{dc}\right]}{dt} = -\frac{\hat{v}_s \Delta \hat{i}_s}{2} \quad (52)$$

Rearranging the terms in (52) gives the required transfer function of the outer voltage-control loop as,

$$Cv_{dc}sv_{dc}(s) = -\frac{\hat{v}_s \hat{i}_s(s)}{2} \quad (53)$$

$$\frac{v_{dc}(s)}{\hat{i}_s(s)} = -\frac{\hat{v}_s}{2sCv_{dc}} \quad (54)$$

The PID controller used for the voltage control loop compensation is described as,

$$G_{v_PID} = K_{vp} + \frac{K_{vi}}{s} + K_{vd}s \quad (55)$$

Therefore, the voltage loop gain is derived as,

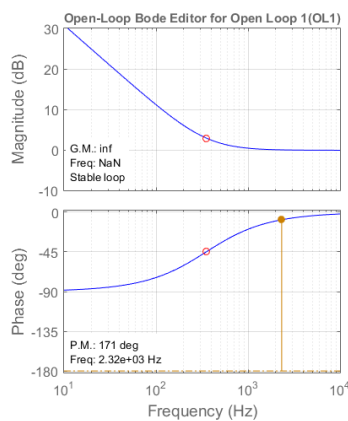


Fig. 21 Bode plot of compensated voltage loop

$$G_{v_PID} \cdot \frac{v_{dc}(s)}{\hat{i}_s(s)} = -\left(K_{vp} + \frac{K_{vi}}{s} + K_{vd}s\right) \times \frac{100.31}{s} \quad (56)$$

A desired loop response with adequate crossover frequency and phase margin is obtained with $K_{vp} = -21.771$ and $K_{vd} = -9.797 \times 10^{-3}$. Fig. 21 shows Bode plot of compensated voltage loop with a cross over frequency of 2.32 kHz and phase margin of 171°. The stability of loop is also ensured.

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REFERENCES

- [1] M. T. A. Khan, G. Norris, R. Chattopadhyay, I. Husain and S. Bhattacharya, "Autoinspection and Permitting With a PV Utility Interface (PUI) for Residential Plug-and-Play Solar Photovoltaic Unit," *IEEE Trans. Ind. Appl.*, vol. 53, no. 2, pp. 1337-1346, March-April 2017.
- [2] M. Montorfano, D. Sbarbaro and L. Morán, "Economic and Technical Evaluation of Solar-Assisted Water Pump Stations for Mining Applications: A Case of Study," *IEEE Trans. Ind. Appl.*, vol. 52, no. 5, pp. 4454-4459, Sept.-Oct. 2016.
- [3] Billel Talbi, Fateh Krim, Toufik Rekioua, Saad Mekhilef, Abdelbasel Laib and Abdesslam Belaout, "A high-performance control scheme for photovoltaic pumping system under sudden irradiance and load changes," *Solar Energy*, vol. 159, pp. 353-368, 2018.
- [4] Packiam Periasamy, N.K. Jain and I.P. Singh, "A review on development of photovoltaic water pumping system," *Renewable and Sustainable Energy Reviews*, vol. 43, pp. 918-925, March 2015.
- [5] R. Kumar and B. Singh, "BLDC Motor Driven Solar PV Array Fed Water Pumping System Employing Zeta Converter," *IEEE Trans. Ind. Appl.*, vol. 52, no. 3, pp. 2315-2322, May-June 2016.
- [6] Bhim Singh and Rajan Kumar, "Solar PV Array Fed Water Pump Driven By BLDC Motor Using Landsman Converter," *IET Renewable Power Generation*, vol. 10, no. 4, pp. 474-484, April 2016.
- [7] K. Rahrah, D. Rekioua, T. Rekioua and S. Bacha, "Photovoltaic pumping system in Bejaia climate with battery storage," *International Journal of Hydrogen Energy*, vol. 40, no. 39, pp. 13665-13675, 19 October 2015.
- [8] Abba Khiareddine, Chokri Ben Salah and Mohamed Faouzi Mimouni, "Power management of a photovoltaic/battery pumping system in agricultural experiment station," *Solar Energy*, vol. 112, pp. 319-338, February 2015.
- [9] Sara Ghaem Sigarchian, Anders Malmquist and Torsten Fransson, "Modeling and Control Strategy of a Hybrid PV/Wind/Engine/Battery System to Provide Electricity and Drinkable Water for Remote Applications," *Energy Procedia*, vol. 57, pp. 1401-1410, 2014.
- [10] A. Khiareddine, C. Ben Salah and M. F. Mimouni, "Strategy of energy control in PVP/battery water pumping system," *International Conference on Green Energy*, Sfax, 2014, pp. 49-54.
- [11] J. H. Ugale and M. S. Panse, "Single phase AC drive for isolated solar photovoltaic water pumping system," *International Conference on Green Computing and Internet of Things (ICGCIoT)*, Noida, 2015, pp. 1285-1287.
- [12] A. Boussaibo, M. Kamta, J. Kayem, D. Toader, S. Haragus and A. Maghet, "Characterization of photovoltaic pumping system model without battery storage by MATLAB/Simulink," *9th International Symposium on Advanced Topics in Electrical Engineering (ATEE)*, Bucharest, 2015, pp. 774-780.
- [13] Zhao Yan, "Photovoltaic Pumping System Using Power Grid as Energy Storage Device," Chinese Patent CN 202455296 U, Sept. 26, 2012.
- [14] Huang, "Photovoltaic Water Pumping and Residual Electricity Grid-Connected System," Chinese Patent CN 204131142 U, Jan. 28, 2015.
- [15] Wang Xing, "High-Efficiency Photovoltaic Pump System," Chinese Patent CN 203884338 U, Octo. 22, 2014.
- [16] Chen steel, Ai Fang, Sun Weilong, Guo Jing and Zhao Xiong, "Photovoltaic Agricultural Power Generating Unit," Chinese Patent CN 203859717 U, Octo. 1, 2014.
- [17] Li Tong qiang, "Solar Power Station with Water Pumping and Energy Storage," Chinese Patent CN 103595337 A, Feb. 19, 2014.
- [18] R. Kumar and B. Singh, "Grid interfaced solar PV based water pumping using brushless DC motor drive," *IEEE International Conference on Power Electronics, Drives and Energy Systems (PEDES)*, Trivandrum, 2016, pp. 1-6.

- [19] *IEEE Recommended Practices and Requirements for Harmonic Control in Electrical Power Systems*, IEEE Standard 519, 1992.
- [20] R. Kumar and B. Singh, "Grid interactive solar PV based water pumping using BLDC motor drive," *IEEE 7th Power India International Conference (PIICON)*, Bikaner, Rajasthan, India, 2016, pp. 1-6.
- [21] A. El Khateb, N. A. Rahim, J. Selvaraj and M. N. Uddin, "Fuzzy-Logic-Controller-Based SEPIC Converter for Maximum Power Point Tracking," *IEEE Trans. Ind. Appl.*, vol. 50, no. 4, pp. 2349-2358, July-Aug. 2014.
- [22] M. Metry, M. B. Shadmand, R. S. Balog and H. Abu-Rub, "MPPT of Photovoltaic Systems Using Sensorless Current-Based Model Predictive Control," *IEEE Trans. Ind. Appl.*, vol. 53, no. 2, pp. 1157-1167, March-April 2017.
- [23] M. A. Elgendy, D. J. Atkinson and B. Zahawi, "Experimental investigation of the incremental conductance maximum power point tracking algorithm at high perturbation rates," *IET Renewable Power Generation*, vol. 10, no. 2, pp. 133-139, Feb. 2016.
- [24] A. Thangavelu, S. Vairakannu and D. Parvathyshankar, "Linear open circuit voltage-variable step-size-incremental conductance strategy-based hybrid MPPT controller for remote power applications," *IET Power Electron.*, vol. 10, no. 11, pp. 1363-1376, Sept. 2017.
- [25] X. Li, H. Wen, L. Jiang, W. Xiao, Y. Du and C. Zhao, "An Improved MPPT Method for PV System With Fast-Converging Speed and Zero Oscillation," *IEEE Trans. Ind. Appl.*, vol. 52, no. 6, pp. 5051-5064, Nov.-Dec. 2016.
- [26] V. Bist and B. Singh, "Reduced sensor configuration of a power factor correction based single-ended primary inductance converter fed brushless DC motor drive," *IET Power Electronics*, vol. 8, no. 9, pp. 1606-1615, Sept. 2015.
- [27] N. Patnaik and A. K. Panda, "Comparative analysis on a shunt active power filter with different control strategies for composite loads," *IEEE Region 10 Conference*, Bangkok, 2014.
- [28] S. Premalatha, Subhransu Sekhar Dash and Paduchuri Chandra Babu, "Power Quality Improvement Features for a Distributed Generation System using Shunt Active Power Filter," *Procedia Engineering*, vol. 64, pp. 265-274, 2013.
- [29] Crystalline Silicon PV Modules [Online]. Available: <https://www.platendt.nl/pdf/bisol-0906-en.pdf>
- [30] N. Mohan, T. M. Undeland and W. P. Robbins, *Power Electronics: Converters, Applications and Design*, 3rd ed. New Delhi, India: John Wiley & Sons Inc., 2010.
- [31] B. Singh, V. Bist, A. Chandra and K. Al-Haddad, "Power Factor Correction in Bridgeless-Luo Converter-Fed BLDC Motor Drive," *IEEE Trans. Ind. Appl.*, vol. 51, no. 2, pp. 1179-1188, March-April 2015.
- [32] R. Kumar Agarwal, I. Hussain and B. Singh, "Three-phase single-stage grid tied solar PV ECS using PLL-less fast CTF control technique," *IET Power Electron.*, vol. 10, no. 2, pp. 178-188, Feb. 2017.
- [33] H. Cai, J. Xiang and W. Wei, "Modelling, analysis and control design of a two-stage photovoltaic generation system," *IET Renewable Power Generation*, vol. 10, no. 8, pp. 1195-1203, Sept. 2016.
- [34] L. Huang, D. Qiu, F. Xie, Y. Chen and B. Zhang, "Modeling and Stability Analysis of a Single-Phase Two-Stage Grid-Connected Photovoltaic System," *Energies*, vol. 10, no. 12, pp. 1-14, 2017.
- [35] V. Bist and B. Singh, "An Adjustable-Speed PFC Bridgeless Buck-Boost Converter-Fed BLDC Motor Drive," *IEEE Trans. Ind. Electron.*, vol. 61, no. 6, pp. 2665-2677, June 2014.
- [36] C. L. Xia, *Permanent Magnet Brushless DC Motor Drives and Controls*. Hoboken, NJ, USA: Wiley, 2012.
- [37] M. A. Elgendy, B. Zahawi and D. J. Atkinson, "Comparison of Directly Connected and Constant Voltage Controlled Photovoltaic Pumping Systems," *IEEE Trans. Sustain. Energy*, vol. 1, no. 3, pp. 184-192, Oct. 2010.
- [38] M. Bhardwaj and S. Choudhury, "Digitally Controlled Solar Micro Inverter Design using C2000 Piccolo Microcontroller," *User's Guide Texas Instruments*, Dallas, Texas, June 2017.
- [39] N. E. Zakzouk, A. K. Abdelsalam, A. A. Helal and B. W. Williams, "PV Single-Phase Grid-Connected Converter: DC-Link Voltage Sensorless Prospective," *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 5, no. 1, pp. 256-546, March 2017.



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